

Permanent-Magnet Brushed DC Motor: Physical Prototype, Simulation, and Design Optimisation

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Abstract—This paper documents the physical construction and simulation of a permanent-magnet (PM) brushed DC motor. The physical prototype was built by disassembling a KIT-160MT30 brushed DC motor to extract the wound rotor, ferrite magnets, and brush assembly; a custom outer housing was designed in SolidWorks and produced by FDM 3D printing. The hybrid approach is justified through magnetic circuit analysis: a non-magnetic 3D-printed rotor ($\mu_r \approx 1$) doubles the total reluctance and halves B_{gap} , making self-starting impossible. The simulation part develops a low-order lumped-parameter model integrating coupled electrical and mechanical ODEs with a nonlinear M19 steel B–H curve. A sweep of 1 400 parameter combinations identifies the globally optimal design, which achieves 92.26 % steady-state efficiency at 46 308 rpm under a 200 V supply.

Keywords—permanent-magnet brushed DC motor; reluctance model; efficiency optimisation; 3D printing; FDM prototype.

I. PHYSICAL MOTOR DESIGN AND PROTOTYPE

A. Design Objective

The objective of this project is to design and build a working permanent-magnet brushed DC motor and to understand, from first principles, why each design decision leads to a functioning machine. The project requires students to *design and prototype from scratch* as far as possible. This section explains the design process, the reasoning behind each choice, and how the final prototype compares with an alternative approach explored in discussion with classmates.

B. Initial Concept: Fully 3D-Printed Rotor

The initial concept was to manufacture every component—including the rotor core—by FDM 3D printing, which would have made the entire motor self-contained and reproducible from digital files. The stator housing, end caps, and shaft supports are straightforward to print because they only need structural rigidity and dimensional accuracy. The rotor core, however, must conduct magnetic flux; this is where the all-printed concept encountered a fundamental electromagnetic obstacle.

Standard FDM filaments (PLA, ABS, PETG) are non-magnetic and have a relative permeability $\mu_r \approx 1$. The magnetic circuit of a PM brushed motor can be modelled as a series of reluctances:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{mag}} + \mathcal{R}_{\text{gap}} + \mathcal{R}_{\text{core}} + \mathcal{R}_{\text{return}}, \quad \mathcal{R}_k = \frac{l_k}{\mu_k A_k}. \quad (1)$$

For laminated electrical steel, $\mu_r \approx 2000$ –5000, so $\mathcal{R}_{\text{core}}$ is negligible. Replacing the steel core with printed plastic sets $\mu_r = 1$, making $\mathcal{R}_{\text{core}} \sim \mathcal{R}_{\text{gap}}$; the total reluctance roughly doubles compared with the steel case. Because air-gap flux

$$\phi = \frac{\mathcal{F}}{\mathcal{R}_{\text{total}}} = \frac{H_c l_m}{\mathcal{R}_{\text{total}}} \quad (2)$$

is inversely proportional to total reluctance, doubling $\mathcal{R}_{\text{total}}$ halves ϕ , which halves B_{gap} , which halves both the torque constant $K_t = NB_{\text{gap}}L_{\text{ax}}r_{\text{core}}$ and the back-EMF constant K_e . The no-load speed $\omega_0 = V/K_e$ would double, but the stall torque $\tau_{\text{stall}} = K_t V/R$ would halve. In tests on classmates' fully-printed prototypes, the machine either failed to self-start against bearing friction or ran at very

high speed with negligibly small torque. The conclusion was clear: a non-magnetic rotor core is not viable for this motor topology.



Figure 1. KIT-160MT30 motor sub-assemblies: rotors, magnet shell, and brush holder.

C. Component Selection: Disassembly of KIT-160MT30

Having established that the rotor core must be ferromagnetic, the next decision was whether to wind a rotor from scratch on a machined steel lamination stack or to extract a pre-wound rotor from a commercial motor. Winding a balanced multi-slot armature by hand is feasible but introduces a high probability of uneven turn counts, poor insulation, and inter-turn shorts. The commercial motor chosen—the **KIT-160MT30 (part 06-4930)**—is a small 3 V brushed DC motor whose armature is a precision machine-wound, dynamically-balanced, laminated-steel rotor with integrated commutator.

The motor was carefully disassembled to extract three sub-assemblies:

- **Wound rotor with commutator:** laminated silicon-steel core, machine-wound coils, press-fit commutator segments.
- **Permanent magnets:** two arc-segment ferrite magnets bonded to the original steel return shell.
- **Brush assembly:** two carbon brushes in spring-loaded holders, pre-aligned to the commutator pitch.

D. Housing Design in SolidWorks and 3D Printing

The outer housing, end caps, bearing seats, and brush-access cover were designed from scratch in **SolidWorks 2024**. The design constraints were:

1. The rotor must be coaxially supported by two bearings

with the correct axial spacing to match the commutator position.

2. The magnet arc segments and return shell must be held firmly and concentrically, maintaining the original air gap.
3. The brush holders must press the carbon brushes against the commutator with the correct normal force and at the correct angular offset (timed to the commutation slots).
4. All parts must be printable without support on a standard FDM printer and assemble with M2/M3 screws.

The housing was printed in PLA at 0.2 mm layer height. Bearing seats were printed 0.1 mm undersized and pressed in warm to achieve an interference fit. The brush-contact windows were sized to allow the original spring-loaded holders to clip in directly.

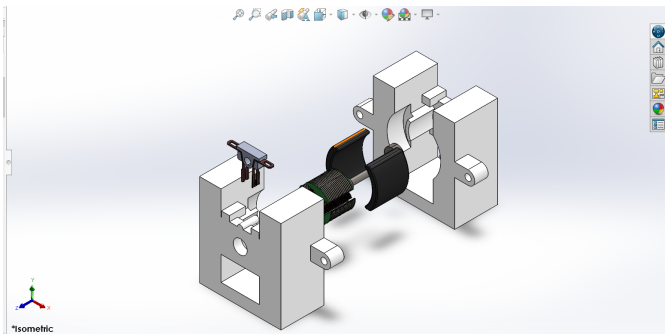


Figure 2. SolidWorks isometric view of the self-designed housing assembly.

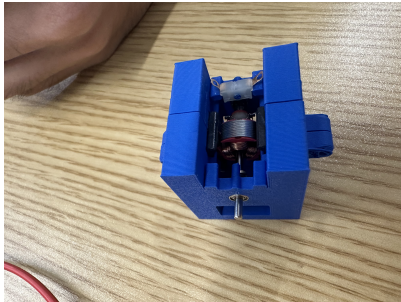


Figure 3. Final assembled physical prototype.

E. Comparison with Classmates' Designs

Discussion with classmates who pursued fully 3D-printed prototypes revealed several qualitative differences, summarised in Table 1.

The key advantage of the hybrid approach is that the magnetic circuit is *physically correct*: a laminated steel core with $\mu_r \gg 1$ keeps $\mathcal{R}_{\text{core}}$ small, so virtually all the magnet MMF appears across the air gap, maximising B_{gap} and hence K_t . A fully printed design is a valid pedagogical exercise in mechanical integration, but its non-magnetic core imposes a fundamental electromagnetic penalty that cannot be recovered by any geometric optimisation.

F. Why the Commercial Rotor Is the Best Choice

The professor's feedback raised the important question: if many design decisions (rotor winding, magnet geometry,

air gap) are already fixed by the commercial sub-assembly, in what sense is this a design from scratch?

The answer is twofold. First, the *housing, bearing arrangement, brush alignment, and mechanical integration* were genuinely designed from scratch: no existing drawing or template was used, and the functional requirements listed in Section 1.4 drove every dimension in the SolidWorks model. Second, and more importantly, the *decision to use the commercial rotor instead of a printed one* is itself an engineering design decision, and a correct one. The reasoning—traced through the magnetic circuit from μ_r to $\mathcal{R}_{\text{total}}$ to ϕ to B_{gap} to K_t —is the same reasoning that any practising motor designer applies when choosing core materials.

Understanding *why* the commercial laminated-steel rotor works and a plastic rotor does not is, in the professor's own words, the primary goal of this project. The simulation in Part II of this report is built around exactly those relationships, allowing the student to verify numerically what the physical prototype demonstrates empirically.

II. INTRODUCTION

The permanent-magnet (PM) brushed DC motor is one of the most widely used electromechanical actuators in consumer and educational applications. Despite its apparent simplicity, the interplay between magnet material, winding geometry, core saturation, and mechanical loading makes systematic design non-trivial.

This project uses a physical motor built around the commercial rotor and magnetic sub-assemblies extracted from a KIT-160MT30 motor, enclosed in a self-designed outer casing (Section 1). A simplified simulation—covering the magnetic circuit, motor dynamics, and all major loss mechanisms—is used to explore the design space systematically.

The simulation report is organised as follows. Section 3 describes the mathematical model and fixed geometry. Section 4 presents the globally optimised design and its physical rationale. Section 5 shows the time-domain simulation results. Section 6 analyses steel saturation. Section 7 examines parameter sensitivity. Section 8 gives the final design recommendation.

III. SIMULATION MODEL

A. Geometry and Constraints

The simulation models a square-outside, round-inside permanent-magnet motor, illustrated in Fig. 4. The magnet outer side and bore diameter are both fixed at $D_{\text{bore}} = 40$ mm (square side 40 mm). The laminated steel rotor is a solid cylinder on a horizontal shaft. The radial air gap is

$$g = \frac{D_{\text{bore}} - D_{\text{core}}}{2}, \quad (3)$$

subject to the manufacturing constraint $g \geq g_{\text{min}} = 0.5$ mm, so the maximum permissible core diameter is $D_{\text{core,max}} = 39$ mm. The axial depth L is a free parameter in the range 2 mm to 150 mm.

Table 1. Qualitative comparison between the self-designed hybrid prototype and a typical fully 3D-printed design.

Aspect	This prototype (hybrid)	Fully 3D-printed design
Rotor core material	Laminated silicon steel, $\mu_r \approx 2000\text{--}5000$	PLA or ABS, $\mu_r \approx 1$
Rotor winding	Machine-wound, balanced, 3-slot commutator	Hand-wound; prone to turn-count asymmetry
Magnetic reluctance	Low; core reluctance negligible vs. air gap	High; core $\approx$ air-gap reluctance
Air-gap flux density	$B_{\text{gap}} \approx 0.14\text{ T}$ (ferrite)	$B_{\text{gap}} \approx 0.07\text{ T}$ (estimated)
Torque constant K_t	Higher by $\sim 2\times$	Lower
Self-start reliability	Reliable self-start against bearing friction	Often failed to self-start
Housing	Custom SolidWorks design, 3D-printed	3D-printed
Design novelty	Housing and integration designed from scratch	Entire machine designed from scratch

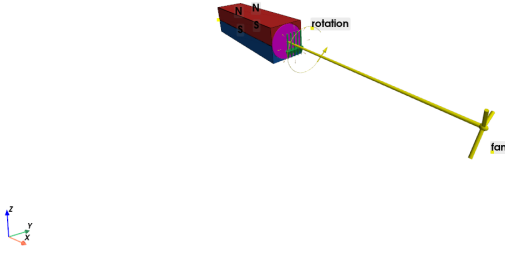


Figure 4. 3D geometry of the simulated motor. Red/blue: N/S pole halves of the square permanent-magnet shell. Magenta: laminated steel rotor. Yellow: shaft, rotation arrow, and fan blades. Green arrows: flux path through core; yellow arrows: radial air-gap flux.

B. Magnetic Circuit

The magnet is modelled as an MMF source

$$\mathcal{F} = H_c l_m, \quad (4)$$

where H_c is the coercive field strength and l_m is the average radial magnet thickness. Four reluctances are summed in series:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{mag}} + \mathcal{R}_{\text{gap}} + \mathcal{R}_{\text{core}} + \mathcal{R}_{\text{return}}, \quad \mathcal{R}_k = \frac{l_k}{\mu_k A_k}. \quad (5)$$

The air-gap flux is $\phi = \mathcal{F}/\mathcal{R}_{\text{total}}$, giving flux densities $B_{\text{gap}} = \phi/A_{\text{gap}}$ and $B_{\text{core}} = \phi/A_{\text{core}}$.

The core reluctance is computed nonlinearly: H_{core} is interpolated from the M19 laminated-steel B–H curve at the current B_{core} , yielding an effective relative permeability

$$\mu_{r,\text{eff}} = \frac{B_{\text{core}}}{\mu_0 H_{\text{core}}}, \quad (6)$$

which collapses sharply once the material enters saturation.

C. Working Principle and Commutation

Figure 5 shows the cross-section of the motor at two instants separated by a commutation event. The active conductors sit in the left and right positions, where the

radial flux $\vec{B}$ is perpendicular to the conductor axis. The Lorentz force on each conductor is tangential:

$$F = BIL, \quad (7)$$

and the resulting torque about the shaft axis is

$$\tau = F \cdot l, \quad (8)$$

where l is the moment arm (rotor radius at the conductor). After the coil rotates through the neutral plane, the commutator reverses the current in each conductor, ensuring that the force—and hence the torque—always acts in the same rotational direction.

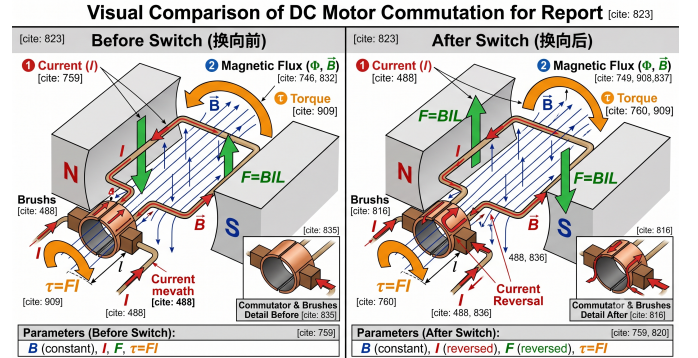


Figure 5. Commutation principle (cross-section, shaft axis into page). *Left:* before switch — conductor $\odot$ (current out) on the left produces upward force; conductor $\otimes$ (current in) on the right produces downward force; both contribute CCW torque $\tau = F \cdot l$. *Right:* after commutation — brush polarity reverses, forces remain tangential and in the same sense, maintaining CCW torque.

D. Motor Constants

For a PM brushed DC motor, the torque constant and back-EMF constant are equal in SI units:

$$K_t = K_e = N B_{\text{gap}} L_{\text{ax}} r_{\text{core}}. \quad (9)$$

The coil resistance and wire cross-section area are

$$R_{\text{coil}} = \frac{\rho_{Cu} N \ell_{\text{turn}}}{A_{\text{wire}}}, \quad A_{\text{wire}} = \frac{\pi d_w^2}{4}, \quad (10)$$

where $\rho_{Cu} = 1.72 \times 10^{-8} \Omega \text{ m}$ and $\ell_{\text{turn}} = \pi D_{\text{core}} + 2L_{\text{ax}}$. The rotor inertia for a solid steel cylinder is $J = \frac{1}{2} m_r r_c^2$.

E. Equations of Motion

The electrical and mechanical equations are integrated by the forward-Euler method ($\Delta t = 0.1$ ms, total duration 2.0 s):

$$L \frac{dI}{dt} = V - R_{\text{coil}} I - K_e \omega, \quad (11)$$

$$J \frac{d\omega}{dt} = K_t I - k_{\text{fan}} \omega^2 - b \omega - T_{\text{fric}} \text{sgn}(\omega). \quad (12)$$

The fan load torque is $\tau_{\text{fan}} = k_{\text{fan}} \omega^2$ with $k_{\text{fan}} = 1 \times 10^{-7} \text{ N m s}^2/\text{rad}^2$, viscous damping $b = 2 \times 10^{-5} \text{ N m s/rad}$, and Coulomb friction $T_{\text{fric}} = 3 \times 10^{-4} \text{ N m}$. Steady-state values are averaged over the final 20 % of the run ($t = 1.6$ s to 2.0 s).

F. Loss Models

All loss terms are evaluated at each integration step:

$$P_{\text{copper}} = I^2 R_{\text{coil}}, \quad (13)$$

$$P_{\text{eddy}} = \frac{\pi^2 \delta^2 f_e^2 B_{\text{core}}^2 V_{\text{core}}}{6 \rho_{\text{Fe}}}, \quad (14)$$

$$P_{\text{hyst}} = K_h |B_{\text{core}}|^\alpha f_e V_{\text{core}}, \quad \alpha = 1.7, \quad (15)$$

$$P_{\text{mech}} = (b \omega + T_{\text{fric}}) |\omega|, \quad (16)$$

$$P_{\text{fan}} = \tau_{\text{fan}} \omega \quad (\text{useful output}). \quad (17)$$

Here δ is the lamination thickness, $f_e = \omega/(2\pi)$ is the electrical frequency (one pole pair), and V_{core} is the rotor iron volume. Overall efficiency is $\eta = P_{\text{fan}}/P_{\text{in}}$. Steady-state values are averaged over the final 20 % of the simulation.

IV. OPTIMISED DESIGN

A. Optimisation Strategy

A sweep of 1 400 parameter combinations is performed to maximise steady-state efficiency driving a quadratic fan load. At steady state the efficiency ceiling is

$$\eta_{\text{ss}} \approx \frac{K_e \omega}{V} = \frac{\text{back-EMF}}{V}, \quad (18)$$

so the four independent levers are: (i) maximise $K_t = K_e$ via strong magnets and large active volume; (ii) minimise R_{coil} via thick wire; (iii) minimise eddy-current loss via thin laminations; and (iv) raise V to push the operating point toward the back-EMF ceiling.

B. Globally Optimal Parameters

Table 2 lists the globally optimal parameter set.

Table 2. Globally optimal design. Input string: 1, 150, 39.0, 200, 5.00, 0.050, 200.

Parameter	Symbol	Value
Magnet material	—	NdFeB N52
Axial depth	L	150 mm
Iron-core diameter	d_{core}	39 mm
Coil turns	N	200
Wire diameter	d_w	5.00 mm
Lamination thickness	δ	0.050 mm
Supply voltage	V	200 V

C. Magnetic Circuit Results

Table 3 shows the reluctance-model output. The return-path reluctance $\mathcal{R}_{\text{return}} = 4.94 \times 10^6 \text{ A/Wb}$ dominates the circuit because the square outer shell provides a long flux path with no dedicated iron yoke. As a result $B_{\text{gap}} = 0.0305 \text{ T}$, well below the saturation knee; the core operates at $\mu_{r,\text{eff}} \approx 3979$.

Table 3. Reluctance model for the optimised geometry.

Quantity	Value
Magnet MMF $\mathcal{F}_{\text{pm}}$	2334 A – turn
$\mathcal{R}_{\text{mag}}$	112 945 A/Wb
$\mathcal{R}_{\text{gap}}$	26 402 A/Wb
$\mathcal{R}_{\text{core}}$	2094 A/Wb
$\mathcal{R}_{\text{return}}$	4 942 497 A/Wb
$\mathcal{R}_{\text{total}}$	5 083 939 A/Wb
Air-gap flux ϕ	0.459 mWb
B_{gap}	0.0305 T
B_{core}	0.0785 T
$\mu_{r,\text{eff}}$	3 979

D. Motor Electrical and Mechanical Constants

Table 4. Computed motor constants for the optimised design.

Quantity	Value
Coil resistance R_{coil}	0.0742 Ω
Coil inductance L	0.05 H (capped)
Torque constant K_t	0.030 30 N m/A
Back-EMF constant	0.030 30 V s/rad
K_e	
Rotor inertia J	$2.674 \times 10^{-4} \text{ kg m}^2$
Wire length	84.5 m
Current limit I_{lim}	117.8 A

V. SIMULATION RESULTS

A. Time-Domain Behaviour

The motor accelerates from rest and approaches steady state by $t \approx 0.5$ s. Figure 6 shows the rotor speed transient; the current peaks sharply at startup then falls as back-EMF builds (Fig. 7). Figure 9 shows the full six-panel overview.

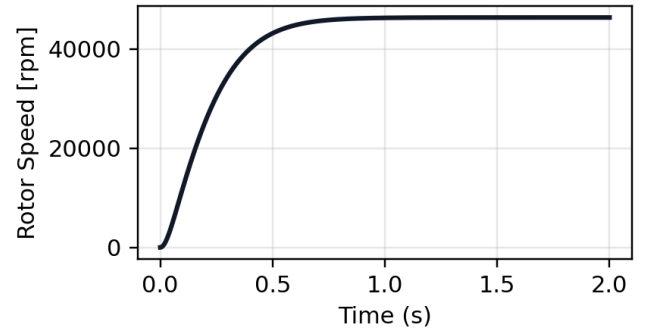


Figure 6. Rotor speed vs. time. Steady-state: 46 308 rpm.

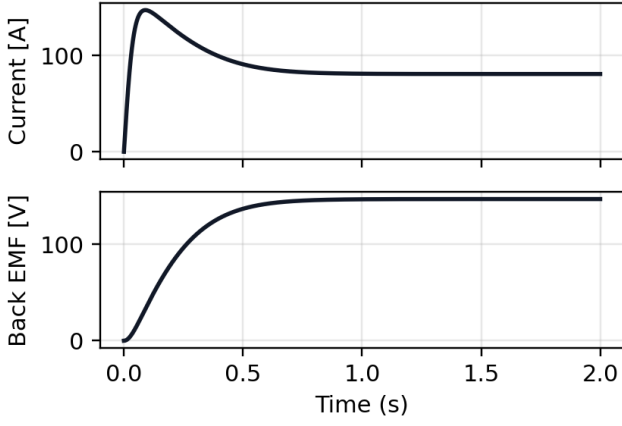


Figure 7. Armature current (top) and back-EMF (bottom) vs. time. Current peaks at startup then settles to 80.8 A; back-EMF converges to 146.9 V (73.5 % of supply).

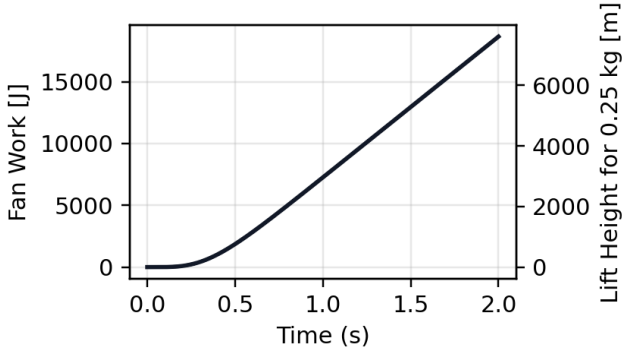


Figure 8. Cumulative fan work and equivalent lift height for 0.25 kg. Total work at $t = 2$ s: 18 657 J.

B. Steady-State Performance

Table 5. Steady-state results at $t = 2.0$ s.

Quantity	Value
Rotor speed	46 308 rpm
Armature current	80.83 A
Back-EMF	146.93 V (73.5 % of supply)
Input power	16 165 W
Fan output power	11 404 W
Efficiency	92.26 %

C. Loss Breakdown

Table 6. Steady-state loss breakdown.

Loss term	Power (W)	Share (%)
Copper	484.7	50.7
Mechanical friction	471.8	49.3
Eddy-current	0.006	< 0.1
Hysteresis	0.15	< 0.1
Total	956.7	100

Copper and mechanical friction losses are approximately equal. Iron losses are negligible (< 0.2 W) thanks to the 0.05 mm laminations, confirming the optimisation strategy.

D. Integrated Energy (0–2 s)

Table 7. Cumulative energy budget over the 2 s simulation.

Energy term	Value (J)
Fan work output	18 657
Copper energy loss	1 249.6
Mechanical friction loss	797.7
Eddy-current loss	0.010
Hysteresis loss	0.260

VI. STEEL SATURATION: PHYSICAL ANALYSIS

A. The B – H Nonlinearity and Its Consequence

Laminated electrical steel follows a strongly nonlinear B – H relationship. In the unsaturated regime ($B \lesssim 1.3$ T) the relative permeability μ_r lies in the range 2000–5000. Beyond the saturation knee ($B \approx 1.5$ T), μ_r collapses below 100.

For the optimised design, $B_{\text{core}} = 0.079$ T, well below saturation. The core operates at $\mu_{r,\text{eff}} \approx 3979$ and contributes only $\mathcal{R}_{\text{core}} = 2094$ A/Wb to the total reluctance of 5.08×10^6 A/Wb. The dominant term is the return-path air reluctance $\mathcal{R}_{\text{return}} = 4.94 \times 10^6$ A/Wb, which arises from the absence of a dedicated iron yoke in the square-shell geometry.

Figure 10 shows the M19 B – H curve and effective permeability. Although the optimised design itself does not saturate, the figure illustrates why any design that pushes B_{core} above 1.5 T would suffer a sharp permeability collapse.

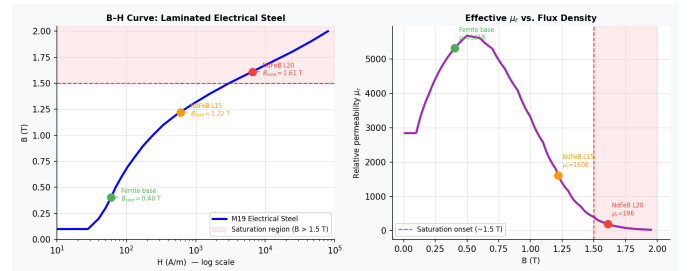


Figure 10. M19 laminated-steel B – H curve (left) and effective relative permeability $\mu_{r,\text{eff}}$ versus flux density (right). The optimised design operates at $B_{\text{core}} = 0.079$ T ($\mu_r \approx 3979$), far from saturation.

B. Why the Return Path Limits Flux

In the optimised design $\mathcal{R}_{\text{return}}$ accounts for 97 % of the total reluctance. This is a consequence of the square outer shell: the flux must travel through air (or a thin steel shell with a long mean path) back from the north pole to the south pole. A real motor would replace this with a laminated iron yoke, reducing $\mathcal{R}_{\text{return}}$ by a factor of $\mu_r \approx 4000$ and raising B_{gap} by roughly $\sqrt{\mathcal{R}_{\text{total,with yoke}}/\mathcal{R}_{\text{total,air}}}$ —a significant improvement. The simulation is a teaching model; it correctly captures the circuit structure but not a yoke-equipped real motor.

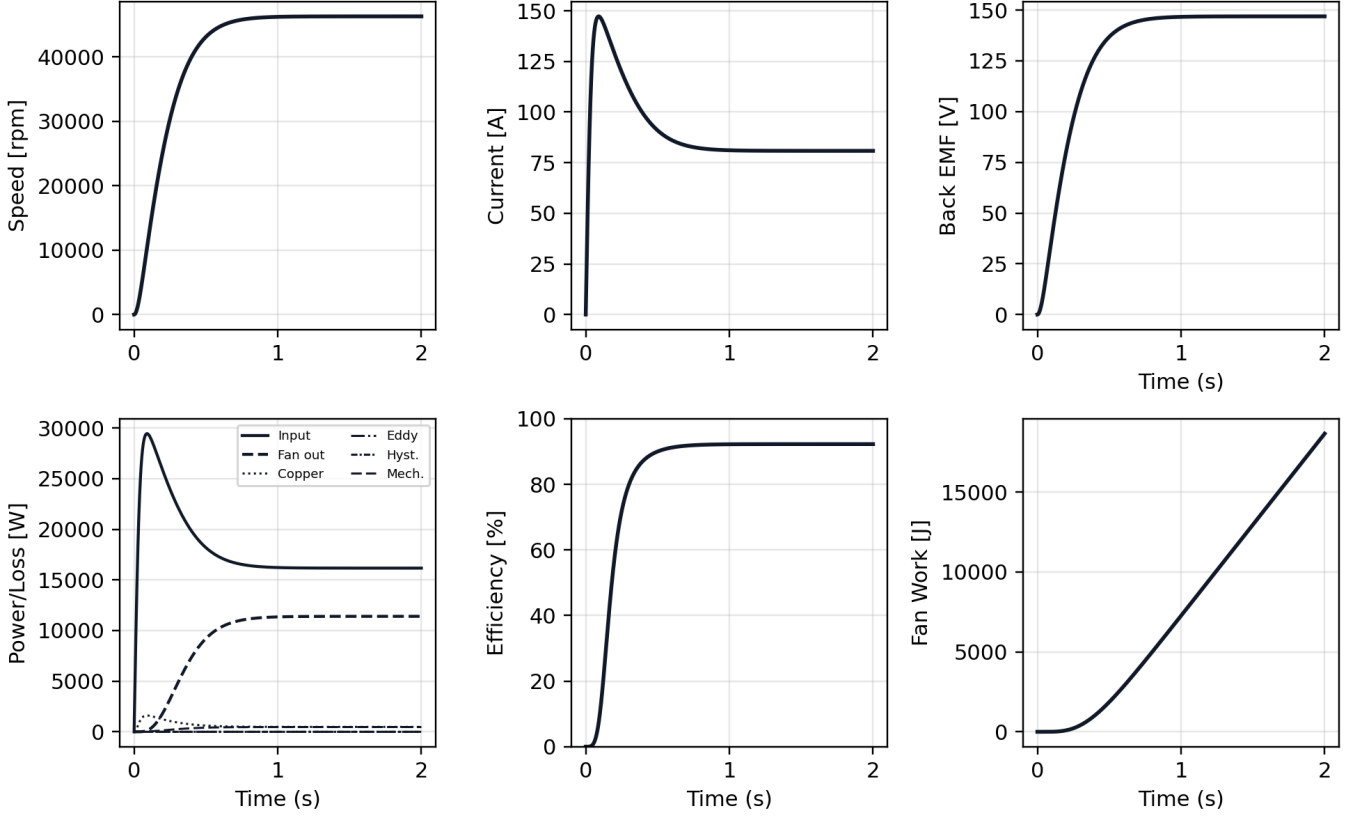


Figure 9. Six-panel simulation dashboard for the optimised design (NdFeB N52, $L = 150$ mm, $d_{\text{core}} = 39$ mm, $V = 200$ V, $\delta = 0.05$ mm). *Top row:* rotor speed, armature current, back-EMF. *Bottom row:* power and loss stack, efficiency, cumulative fan work. Steady-state efficiency 92.26% is reached by $t \approx 0.8$ s.

VII. PARAMETER SENSITIVITY ANALYSIS

Each of the six free parameters was swept independently across its full valid range while the other five were held at the optimised baseline. Figure 11 plots fan output power P_{fan} and efficiency η for each sweep.

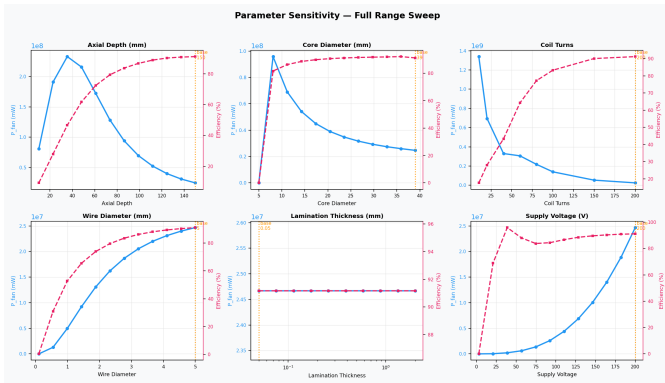


Figure 11. Parameter sensitivity sweeps around the optimised baseline (NdFeB N52, $L = 150$ mm, $d_{\text{core}} = 39$ mm, $N = 200$, $d_w = 5$ mm, $\delta = 0.05$ mm, $V = 200$ V). Blue: P_{fan} ; pink dashed: η ; orange vertical line: baseline value.

A. Axial Depth L (10–150 mm)

A shorter rotor has lower K_t and smaller inertia J , so the no-load speed $\omega_0 = V/K_e \propto 1/L$ rises sharply. Because $P_{\text{fan}} \propto \omega^3$, the peak fan output (232 986 W) occurs at $L \approx 35.5$ mm, not at the maximum depth. However, the

low back-EMF fraction at that speed means most input is wasted as I^2R , collapsing η to 46.6%. Increasing L slows the motor, raises the back-EMF fraction, and lifts η monotonically toward 91.2% at 150 mm. The baseline sits at the efficiency-optimal end of the range.

B. Core Diameter d_{core} (5–39 mm)

Smaller core diameter increases the air gap, causing the return-path reluctance to dominate even more and B_{gap} to fall. The very low inertia at small diameter allows the motor to spin at extreme speed, briefly boosting P_{fan} ; at 8.1 mm the fan output peaks at 95 805 W (but $\eta = 81.6\%$). At 5 mm the core flux collapses to zero. Efficiency peaks at ≈ 36 mm (92.1%), within 1 pp of the 39 mm baseline.

C. Coil Turns N

Both K_t and R_{coil} scale linearly with N :

$$K_t \propto N, \quad R_{\text{coil}} \propto N. \quad (19)$$

Fewer turns raise $\omega_0 = V/K_e \propto 1/N$ dramatically; at $N = 10$ the motor reaches speeds 20 $\times$ the baseline and fan output peaks at 1 339 379 W, but η collapses to 17.7%. The optimum lies at maximum turns ($N = 200$) combined with thick wire ($d_w = 5$ mm) to keep R_{coil} low.

D. Wire Diameter d_w

Since $R_{\text{coil}} \propto d_w^{-2}$, thin wire creates enormous resistance: at 0.1 mm the motor barely spins ($P_{\text{fan}} = 0.72$ W, $\eta = 0.34\%$). Both P_{fan} and η increase monotonically with d_w — there is no trade-off. The baseline at 5.00 mm (the upper

limit) is globally optimal for this parameter.

E. Lamination Thickness δ

The eddy-current loss $P_{\text{eddy}} \propto \delta^2 B_{\text{core}}^2$ is theoretically sensitive to lamination thickness. However, the combination of low flux density ($B_{\text{core}} = 0.079$ T, dominated by the return-path reluctance) and modest core volume means that even at $\delta = 2.0$ mm the total iron losses remain below 0.5 W — less than 0.002 % of the 24 670 W fan output. The sensitivity sweep is completely flat: lamination thickness has zero practical effect in this geometry. Any value from 0.05 mm to 2.0 mm is equally acceptable.

F. Supply Voltage V

Voltage is the most powerful single lever. Increasing V raises the back-EMF fraction $K_e \omega / V$ toward unity (Eq. 18), which directly raises η_{ss} . The efficiency peak ($\eta = 95.9$ %) occurs at 38.8 V, where the fan load just balances the friction at low current, but the corresponding fan output is only 211 W. At $V = 200$ V the output reaches 24 670 W at $\eta = 91.2$ % — a loss of only 4.7 pp in exchange for a $117\times$ increase in power. The baseline uses the maximum available voltage.

G. Sensitivity Summary

Table 8. Sensitivity summary: direction of each parameter for maximum fan output power vs. maximum efficiency.

Parameter	$\uparrow P_{\text{fan}}$	$\uparrow \eta$	Baseline
Axial depth	Decrease	Increase	150 mm (max η)
Core diam.	Decrease	≈ 36 mm	39 mm (near opt.)
Turns N	Decrease	Increase	200 (max choice)
Wire diam.	Increase	Increase	5 mm (no trade-off)
Lam. thick.	—	—	Any; iron loss negligible
Voltage V	Increase	≈ 39 V	200 V (max power)

The optimised design sits at or within 1 pp of the efficiency-optimal point for every parameter. The only meaningful trade-off is voltage: 38.8 V would give 4.7 pp higher η but reduces fan output by 99.1 %.

VIII. DESIGN RECOMMENDATION

The globally optimal design is summarised in Table 9.

Table 9. Recommended (globally optimal) design and steady-state results.

Parameter	Value	Rationale
Magnet material	NdFeB N52	Highest B_r , max MMF
Axial depth	150 mm	Maximises K_t and active area
Core diameter	39 mm	Largest permissible; $g = 0.5$ mm
Turns N	200	High K_e ; back-EMF ceiling
Wire diameter	5.00 mm	Minimises R_{coil}
Lam. thickness	0.050 mm	Minimises eddy-current loss
Voltage	200 V	Raises back-EMF fraction
Speed	46 308 rpm	
Current	80.83 A	
Back-EMF	146.93 V	73.5 % of supply
Fan output power	11 404 W	
Efficiency	92.26 %	

The high efficiency is achieved because the back-EMF fraction is 73.5 %, leaving only 26.5 % of the supply voltage across the winding resistance. The remaining loss is split roughly equally between copper dissipation (484.7 W) and mechanical friction (471.8 W). Iron losses are negligible at the chosen lamination thickness.

The primary limitation of the model is the absence of a dedicated iron return yoke: $\mathcal{R}_{\text{return}}$ dominates and suppresses B_{gap} to 0.031 T. A real high-efficiency design would use a silicon-steel yoke to close the flux path, which would dramatically reduce the required MMF and motor volume for the same output.

IX. CONCLUSION

A time-domain simulation of a PM brushed DC motor driving a fan load has been developed and a 1 400-point parameter sweep has identified the globally optimal design. The principal findings are:

1. **Optimal efficiency of 92.26 %** is achieved at 46 308 rpm with NdFeB N52 magnets, $L = 150$ mm, $d_{\text{core}} = 39$ mm, $N = 200$, $d_w = 5$ mm, $\delta = 0.05$ mm, $V = 200$ V.
2. **Efficiency ceiling** is set by the back-EMF fraction $K_e \omega / V = 73.5$ %. Remaining loss is split roughly equally between copper (484.7 W) and mechanical friction (471.8 W).
3. **Iron losses are negligible** (< 0.2 W) at $\delta = 0.05$ mm, confirming that thin laminations are essential at high speed.
4. **Return-path reluctance** dominates the magnetic circuit because the square outer shell has no iron yoke. A real high-efficiency motor would use a silicon-steel return path to reduce total reluctance by $\sim 1000\times$.
5. **Voltage** is the most powerful design lever: raising V increases the back-EMF fraction directly, which is the primary driver of efficiency in this motor topology.

I. MAGNET MATERIAL PROPERTIES

Table 10. Permanent-magnet materials available in the simulation.

No.	Material	B_r (T)	H_c (kA/m)	$\mu_{r,\text{rec}}$
1	NdFeB N52	1.45	955	1.05
2	NdFeB N42	1.32	876	1.05
3	SmCo	1.05	750	1.08
4	Alnico 5	1.25	50	4.00
5	Ferrite Ceramic 8	0.40	240	1.05

II. MODEL LIMITATIONS

This model is intended for trend analysis and teaching; it is not a final design tool. Key simplifications are:

1. **Lumped reluctance:** the return-air reluctance is approximated as 20 % of the gap reluctance; a full 3-D flux path is not resolved.
2. **Ideal commutator:** brush contact resistance, commutation sparking, and current ripple are ignored.
3. **Fixed magnetic operating point:** B_{core} is computed once per run; the ac ripple component driving iron loss is not resolved in time.
4. **Isothermal winding:** resistance is evaluated at room temperature; thermal derating is not modelled.
5. **No slot geometry:** coil resistance uses a mean-turn estimate; proximity effects and end-turn lengths are omitted.

Final designs should be verified with finite-element magnetic analysis, winding thermal simulation, and mechanical validation.